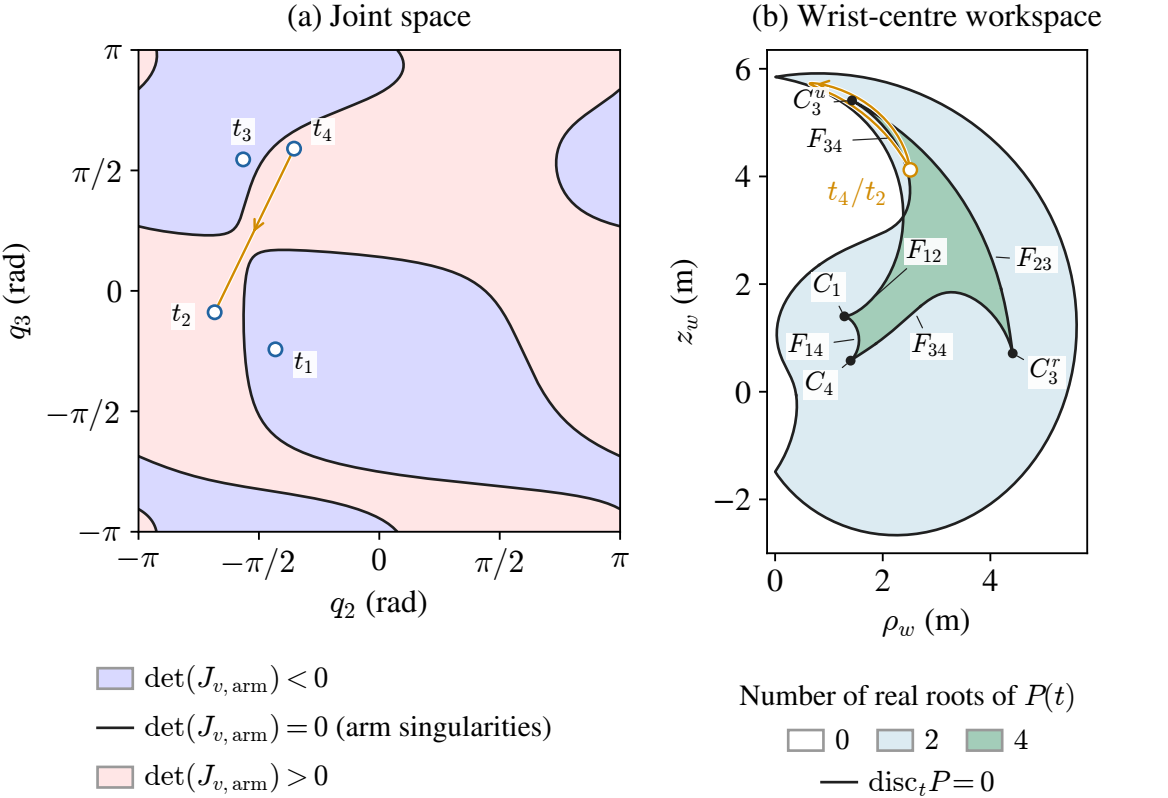

(a) Joint space
(b) Wrist-centre workspace
q_3 (rad)
q_2 (rad)
t_1
t_2
t_3
t_4
z_w (m)
ρ_w (m)
C_3^u
F_{34}
t_4/t_2
F_{12}
F_{23}
C_1
F_{14}
C_4
C_3^r
$\det(J_{v,\mathrm{arm}}) < 0$
$\det(J_{v,\mathrm{arm}}) = 0$ (arm singularities)
$\det(J_{v,\mathrm{arm}}) > 0$
Number of real roots of $P(t)$
0
2
4
$\mathrm{disc}_t P = 0$